Censorship-Resistant Mesh Networking for Diaspora Communities

Pars Network Technical Whitepaper PNP-002

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Abstract

We present Pars Mesh, a censorship-resistant mesh networking protocol designed for diaspora communities operating under heterogeneous network conditions ranging from unrestricted internet access to complete national firewall environments. The protocol operates across three tiers: a direct peer-to-peer overlay for internet-based routing, a local radio mesh for proximity connectivity, and integration adapters for Tor and I2P anonymous networks. We introduce a relay incentive mechanism based on bandwidth proofs that compensates operators through Pars L2 micropayments without revealing routing information. The protocol includes mobile-optimised mesh capabilities using Bluetooth Low Energy and Wi-Fi Direct for scenarios where internet connectivity is entirely unavailable. NAT traversal is achieved through STUN, TURN, and ICE with protocol obfuscation to resist deep packet inspection. Pars Mesh achieves median message delivery latency of 1.8 s under normal conditions and 12.4 s under simulated national firewall conditions, with a 99.2% delivery rate across all tested scenarios.

Keywords: mesh networking, censorship resistance, relay incentives, NAT traversal, mobile mesh, diaspora communication

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1 Introduction

Internet censorship represents one of the most significant threats to diaspora communities. The Iranian internet infrastructure employs a multi-layered censorship apparatus: DNS poisoning, IP blocking, deep packet inspection (DPI), and periodic internet shutdowns [1]. During the 2019 and 2022 protests, authorities implemented near-total blackouts lasting days to weeks [2].

For diaspora communities censorship creates a dual problem: members within censored jurisdictions cannot access services, and members outside cannot reliably communicate with those inside. Traditional circumvention tools (VPNs, Tor) depend on internet connectivity existing in the first place.

1.1 Design Requirements

R1: Multi-path delivery: Messages reach recipients through at least two independent network paths.

R2: Graceful degradation: Operation continues as conditions deteriorate from unrestricted to fully disconnected.

R3: Protocol obfuscation: Traffic indistinguishable from benign HTTPS under DPI.

R4: Incentive alignment: Relay operators compensated for bandwidth without compromising routing privacy.

R5: Mobile-first: Operates on mobile devices with constrained resources.

1.2 Contributions

- Three-tier mesh architecture that degrades gracefully from internet overlay to local Bluetooth mesh.
- Bandwidth proof mechanism for relay incentives preserving routing privacy.
- Protocol obfuscation defeating state-of-the-art DPI.
- Mobile mesh optimised for BLE and Wi-Fi Direct.
- Evaluation under simulated censorship modelled after real-world national firewalls.

2 Threat Model

2.1 Adversary Capabilities

Definition 2.1 (State-Level Adversary)**.** *The adversary $\mathcal{A}$ controls:*

(a) All network infrastructure within jurisdiction J

(b) DPI systems capable of real-time protocol fingerprinting

(c) Ability to block arbitrary IP addresses and port ranges

(d) Ability to throttle or disable internet within J

(e) Limited ability to compel cooperation from foreign operators

2.2 Security Goals

1. **Availability:** ≥ 0.99 delivery when at least one tier is operational.
2. **Confidentiality:** End-to-end encryption not accessible to $\mathcal{A}$.
3. **Unlinkability:** $\mathcal{A}$ cannot determine sender-recipient pairs with non-negligible advantage.
4. **Undetectability:** Pars Mesh traffic computationally indistinguishable from allowed HTTPS under DPI.

3 Three-Tier Mesh Architecture

3.1 Overview

Table 1: Mesh Tier Characteristics

Tier	Transport	Range	Latency
T1: Internet Overlay	TCP/QUIC	Global	50–500 ms
T2: Local Mesh	BLE / Wi-Fi Direct	10–100 m	10–100 ms
T3: Bridge Relay	Tor / I2P / Meek	Global	1–10 s

Algorithm 1 Adaptive Tier Selection

```
function SelectTier(destination)
    if InternetOverlay.IsReachable(destination) then
        return Tier1
    end if
    if BridgeRelay.HasRoute(destination) then
        if LocalMesh.HasRoute(destination) then
            return arg min_latency(T2, T3)
        end if
        return Tier3
    end if
    if LocalMesh.HasRoute(destination) then
        return Tier2
    end if
    Enqueue for deferred delivery
end function
```

3.2 Tier 1: Internet Overlay

The primary path when unrestricted internet is available. Uses a Kademlia-based [3] DHT for peer discovery.

3.2.1 XOR Distance Metric

Definition 3.1 (XOR Distance)**.** *For node IDs $a, b \in \{0,1\}^{256}$: $d(a,b) = a \oplus b$. Nodes are placed in k-buckets where bucket i contains nodes with $2^i \leq d < 2^{i+1}$.*

3.2.2 Transport Protocol

Tier 1 uses QUIC [4]:

- Indistinguishable from HTTPS/3 under DPI.
- Built-in connection migration handles NAT rebinding and network switches on mobile.
- Multiplexed streams avoid head-of-line blocking.
- 0-RTT establishment reduces latency for repeat connections.

3.2.3 Domain Fronting

Listing 1: Domain Fronting Configuration

```go
type DomainFrontConfig struct {
    OuterDomain string   // SNI for DPI
    InnerDomain string   // actual Host header
    CDNEndpoint string   // CDN edge address
    Fallbacks   []string // blocked-domain fallbacks
}

var defaultFronts = []DomainFrontConfig{
    {
        OuterDomain: "cdn.example.com",
        InnerDomain: "relay.pars.network",
        CDNEndpoint: "cdn-edge.example.com:443",
        Fallbacks:   []string{
            "static.example.org",
            "assets.example.net",
        },
    },
}
```

3.3 Tier 2: Local Mesh

Provides connectivity when internet access is unavailable or too dangerous.

3.3.1 BLE Gossip Protocol

Algorithm 2 BLE Mesh Gossip Protocol

```
function BLEGossip(message m)
    id ← H(m)
    if id ∈ seen then
        return                                  ▷ deduplicate
    end if
    seen.Add(id, TTL)
    ProcessMessage(m)
    if m.TTL > 1 then
        m.TTL ← m.TTL − 1
        for all p ∈ BLENeighbours do
            BLESend(p, m)                       ▷ async
        end for
    end if
end function
```

Definition 3.2 (BLE Fragment)**.** *A fragment F_i is:* $\langle \text{msgId}[4], \text{fragIdx}[1], \text{fragTotal}[1], \text{payload}[\leq 25] \rangle$*, consuming 31 bytes. A 1 KB message requires* $\lceil 1024/25 \rceil = 41$ *fragments.*

3.3.2 Wi-Fi Direct Groups

Listing 2: Wi-Fi Direct Group Management

```go
type WiFiDirectMgr struct {
    owner     bool
    peers     map[PeerID]*WDPeer
    maxSize   int
    channel   int
}

func (m *WiFiDirectMgr) FormGroup(
    ctx context.Context,
) error {
    intent := m.goIntent()
    result, err := m.negotiate(ctx, intent)
    if err != nil {
        return fmt.Errorf("GO negotiate: %w", err)
    }
    m.owner = result.IsGroupOwner
    if m.owner {
        return m.startOwner(ctx)
    }
    return m.join(ctx, result.GroupAddr)
}

func (m *WiFiDirectMgr) goIntent() int {
    bat := getBatteryLevel()
    switch {
    case bat > 80: return 14
    case bat > 40: return 8
    default:       return 2
    }
}
```

3.4 Tier 3: Bridge Relay

Integrates with Tor, I2P, and circumvention systems (Meek, Snowflake).

3.4.1 Tor Integration

Listing 3: Tor Hidden Service Setup

```go
type TorBridge struct {
    socksAddr string
    hsDir     string
    useObfs4  bool
    bridges   []string
    cmd       *exec.Cmd
}

func NewTorBridge(cfg TorBridgeConfig) (
    *TorBridge, error,
) {
    torrc := buildTorrc(cfg)
    cmd := exec.Command("tor", "-f", torrc)
    if err := cmd.Start(); err != nil {
        return nil, fmt.Errorf("tor start: %w", err)
    }
    if err := waitBootstrap(
        cfg.SOCKSAddr, 120*time.Second,
    ); err != nil {
        cmd.Process.Kill()
        return nil, fmt.Errorf("tor boot: %w", err)
    }
    return &TorBridge{
        socksAddr: cfg.SOCKSAddr,
        hsDir:     cfg.HiddenServiceDir,
        useObfs4:  cfg.UseObfs4,
        bridges:   cfg.Bridges,
        cmd:       cmd,
    }, nil
}
```

3.4.2 I2P Integration

Listing 4: I2P Tunnel via SAM v3.1

```go
type I2PConfig struct {
    SAMAddr       string
    TunnelLen     int
    TunnelQty     int
    InboundNick   string
    OutboundNick  string
}

func (c *I2PConfig) CreateSession(
    ctx context.Context,
) (*I2PSession, error) {
    conn, err := net.Dial("tcp", c.SAMAddr)
    if err != nil {
        return nil, fmt.Errorf("SAM: %w", err)
```

```
    }
    fmt.Fprintf(conn,
        "HELLO VERSION MIN=3.1 MAX=3.1\n")
    fmt.Fprintf(conn,
        "SESSION CREATE STYLE=STREAM "+
        "ID=%s DESTINATION=TRANSIENT "+
        "inbound.length=%d outbound.length=%d "+
        "inbound.quantity=%d outbound.quantity=%d\n",
        c.InboundNick,
        c.TunnelLen, c.TunnelLen,
        c.TunnelQty, c.TunnelQty,
    )
    return parseSessionReply(conn)
}
```

4 NAT Traversal

4.1 Problem Statement

Approximately 70% of mobile devices and 40% of desktops operate behind NAT [5], preventing direct peer connections.

4.2 ICE Framework

Algorithm 3 Extended ICE Connectivity Check

```
function ICEConnect(local, remote)
    C ← ∅
    C ← C ∪ HostCandidates()
    C ← C ∪ STUNCandidates()
    C ← C ∪ TURNCandidates()
    C ← C ∪ ObfuscatedCandidates()
    SortByPriority(C)
    for all c ∈ C do
        p ← FormPair(c, remote.cands)
        if ConnectivityCheck(p) then
            return p
        end if
    end for
    return ConnectViaTURN(local, remote)
end function
```

4.3 Obfuscated STUN

Definition 4.1 (Obfuscated STUN)**.** *An obfuscated STUN message M' is derived from standard message M by $M' = E_k(M) \| \mathrm{HMAC}_k(E_k(M))$ where k is derived from the relay's public key and E_k is AES-128-CTR.*

4.4 NAT Traversal Success Rates

For symmetric NAT pairs (common in cellular networks) the system falls back to TURN relay.

Table 2: Hole-Punching Success Rates

NAT Pair	UDP	TCP
Full Cone – Full Cone	99.8%	98.2%
Full Cone – Restricted	98.1%	95.7%
Restricted – Restricted	94.3%	89.1%
Port Restricted – Port Restr.	72.6%	61.3%
Symmetric – Any	31.2%	18.7%
Symmetric – Symmetric	8.4%	3.2%

5 Relay Incentive Mechanism

5.1 Design Goals

1. **Verifiability:** Relay work verifiable without revealing content or routing.
2. **Proportionality:** Compensation proportional to actual bandwidth contributed.
3. **Sybil Resistance:** Multiple identities yield no extra reward beyond actual bandwidth.

5.2 Bandwidth Proofs

Definition 5.1 (Bandwidth Proof)**.** $\Pi = \langle R, e, n, \{h_i\}_{i=1}^{n}, \sigma_R \rangle$ *where* n *is the relay operation count,* $h_i = H(\mathrm{msg}_i \| R \| e)$*, and* σ_R *is the relay's signature.*

Algorithm 4 Bandwidth Proof Verification

Require: Proof Π from relay R, challenge set $\mathcal{C}$

$v \leftarrow 0$
for all $i \in \mathcal{C}$ **do**
  if $S_i.\mathrm{Confirm}(h_i) \wedge D_i.\mathrm{Confirm}(h_i)$ **then**
    $v \leftarrow v + 1$
  end if
end for
$\rho \leftarrow v/|\mathcal{C}|$
if $\rho \geq 0.9$ **then**
  Pay R: $n \cdot r_{\mathrm{msg}} \cdot \rho$ via L2
else
  Flag R for investigation
end if

5.3 Payment Channels

Listing 5: Relay Payment Channel

```
contract RelayPaymentChannel {
    struct Chan {
        address sender;
        address relay;
        uint256 deposit;
        uint256 expiry;
        uint256 nonce;
    }
```

```
    mapping(bytes32 => Chan) public chans;

    function open(
        address relay, uint256 dur
    ) external payable {
        bytes32 id = keccak256(
            abi.encodePacked(
                msg.sender, relay, block.number
            )
        );
        chans[id] = Chan(
            msg.sender, relay,
            msg.value, block.timestamp + dur, 0
        );
    }

    function close(
        bytes32 id, uint256 amt,
        uint256 nonce, bytes calldata sig
    ) external {
        Chan storage c = chans[id];
        require(msg.sender == c.relay);
        require(nonce > c.nonce);
        require(amt <= c.deposit);
        bytes32 h = keccak256(
            abi.encodePacked(id, amt, nonce)
        );
        require(
            ECDSA.recover(h, sig) == c.sender
        );
        c.nonce = nonce;
        payable(c.relay).transfer(amt);
        payable(c.sender).transfer(
            c.deposit - amt
        );
        delete chans[id];
    }
}
```

5.4 Sybil Resistance

Definition 5.2 (Relay Stake)**.** *Relay R claiming bandwidth B (MB/s) must stake $S_R = \max(S_{\min},\ \alpha B)$ where $S_{\min} = 1000$ PARS and $\alpha = 100$ PARS/MB/s.*

False bandwidth proofs trigger proportional slashing.

Table 3: Mobile Platform Constraints

Constraint	iOS	Android
BLE advertising (bg)	Limited	Supported
BLE scanning (bg)	Limited	Supported
Wi-Fi Direct	Not supported	Supported
Background execution	30 s max	Foreground svc
Wake-on-BLE	Supported	Supported
Max BLE connections	7–10	7–15

6 Mobile Mesh Protocol

6.1 Platform Constraints

6.2 Battery-Aware Scheduling

Algorithm 5 Battery-Aware Mesh Scheduling

```
function SCHEDULE(batteryLevel)
    if batteryLevel > 80% then
        BLE scan ← 1 s;  relay ← true;  WD ← on
    else if batteryLevel > 40% then
        BLE scan ← 5 s;  relay ← false;  WD ← on-demand
    else if batteryLevel > 15% then
        BLE scan ← 30 s;  relay ← false;  WD ← off
    else
        BLE scan ← off;  receive-only via BLE advertising
    end if
end function
```

6.3 Store-and-Forward

When recipients are unreachable, epidemic routing [8] is used:

Definition 6.1 (Epidemic Message)**.** $M = \langle \text{id}, \text{dst}, \text{payload}, \text{TTL}, \text{hops}, \text{created}, \text{priority} \rangle$. *Each node stores M and forwards to all encountered peers not holding it, until TTL expires or delivery is confirmed.*

Proposition 6.1 (Delivery Probability)**.** *With n nodes and encounter rate λ: $P(\text{delivery by } t) = 1 - e^{-\lambda(n-1)t}$.*

6.4 Proximity Group Formation

Listing 6: Proximity Group

```go
type ProximityGroup struct {
    ID        GroupID
    Members   map[PeerID]*BLEPeer
    RSSI      map[PeerID]int
    Created   time.Time
    GroupKey  [32]byte
}

func (g *ProximityGroup) ShouldMerge(
```

```go
    other *ProximityGroup,
) bool {
    overlap := 0
    for id := range g.Members {
        if _, ok := other.Members[id]; ok {
            overlap++
        }
    }
    threshold := min(
        len(g.Members), len(other.Members),
    ) / 2
    return overlap >= threshold
}

func (g *ProximityGroup) RotateKey() error {
    epoch := time.Now().Unix() / 3600
    newKey := hkdf.Extract(sha256.New,
        g.GroupKey[:],
        []byte(fmt.Sprintf("epoch:%d", epoch)),
    )
    copy(g.GroupKey[:], newKey)
    return g.distributeKey()
}
```

7 Protocol Obfuscation

7.1 DPI Resistance

DPI identifies protocols through:

1. Port-based fingerprinting
2. Byte-pattern signature matching
3. Statistical flow analysis (packet sizes, timing)

7.2 Obfs4-Inspired Transport

Inspired by obfs4 [9]:

- Uniform random byte distribution in payloads (defeats signature DPI)
- Randomised packet lengths from a configurable distribution (defeats statistical DPI)
- NTor [10] key exchange with TLS-like handshake pattern

Algorithm 6 Obfuscated Handshake

Require: Client C, server S with public key pk_S

C: generate $(x, X = xG)$
mark $\leftarrow \mathrm{HMAC}(\mathrm{pk}_S, X)[:16]$
$C \rightarrow S$: $X \| \mathrm{pad} \| \mathrm{mark} \| \mathrm{MAC}$
S: scan for mark to locate X
S: generate $(y, Y = yG)$
$\mathrm{ss} \leftarrow H(xY \| X \cdot \mathrm{sk}_S)$
$S \rightarrow C$: $Y \| \mathrm{auth} \| \mathrm{pad}$
Both derive $(k_1, k_2) = \mathrm{KDF}(\mathrm{ss})$

7.3 Traffic Shaping

Listing 7: Traffic Shaping Engine

```go
type TrafficProfile struct {
    SizeCDF  []float64
    SizeBins []int
    IATMean  time.Duration
    IATStd   time.Duration
    BurstP   float64
    BurstN   int
}

var HTTPSProfile = TrafficProfile{
    SizeCDF:  []float64{
        0.1, 0.3, 0.5, 0.7, 0.85, 0.95, 1.0,
    },
    SizeBins: []int{
        64, 128, 256, 512, 1024, 1380, 1460,
    },
    IATMean: 50 * time.Millisecond,
    IATStd:  30 * time.Millisecond,
    BurstP:  0.15,
    BurstN:  8,
}
```

8 Evaluation

8.1 Scenarios

Table 4: Evaluation Scenarios

Scenario	Internet	DPI	Blocking
S1: Normal	Full	None	None
S2: Filtered	Full	Active	Protocol
S3: Throttled	128 kbps	Active	IP + Protocol
S4: Blackout	None	N/A	Total

8.2 Results

Table 5: Message Delivery Performance

Scenario	Delivery	p50 Latency	p99 Latency
S1	99.97%	0.4 s	2.1 s
S2	99.81%	1.8 s	8.3 s
S3	99.23%	4.2 s	31.7 s
S4	87.41%	12.4 s	180 s

S4 delivery depends on the local mesh. 87.4% reflects simulation with 100 nodes/km^2 in an urban setting.

8.3 DPI Evasion

Obfuscated traffic classified as "HTTPS/TLS" by commercial DPI in 98.7% of flows; remaining 1.3% classified as "Unknown/Encrypted" (not typically blocked).

8.4 Battery Impact

Table 6: Battery Consumption (24 h, 4500 mAh device)

Mode	Consumed
FULL	18.3%
NORMAL	8.7%
LOW_POWER	3.2%
CRITICAL	0.9%

9 Related Work

Mesh Systems. Briar [11] provides P2P encrypted messaging with Tor and BLE. Bridgefy [12] offers BLE mesh with weaker encryption. GoTenna [13] uses dedicated radio. Pars Mesh combines all tiers with economic incentives and blockchain integration.

Circumvention. Tor [14] and pluggable transports (obfs4 [9], meek [15], Snowflake [16]) improve DPI resistance. Psiphon [17] combines multiple techniques. Pars Mesh integrates these as Tier 3 fallbacks.

Incentivised Relays. Orchid [18] (decentralised VPN), HOPR [19] (incentivised mixnet), Nym [20] (privacy infrastructure). Pars Mesh's bandwidth proofs provide stronger Sybil resistance via sender/recipient co-signing.

10 Conclusion

Pars Mesh provides censorship-resistant mesh networking for diaspora communities. The three-tier architecture enables graceful degradation from internet overlay to local BLE mesh. Bandwidth proofs align relay incentives with privacy. Protocol obfuscation defeats DPI. Evaluation demonstrates 99.2% delivery across diverse censorship scenarios and 87.4% during complete blackouts. Battery consumption remains below 9% in normal operation.

Future work includes satellite integration for extreme scenarios, adaptive ML-based traffic shaping, and formal game-theoretic analysis of relay incentives.

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A Wire Protocol Message Formats

Listing 8: Wire Protocol Constants

```go
const (
    MsgPing           = 0x01
    MsgPong           = 0x02
    MsgFindNode       = 0x03
    MsgFoundNode      = 0x04
    MsgStore          = 0x05
    MsgStoreAck       = 0x06
    MsgRelay          = 0x07
    MsgRelayAck       = 0x08
    MsgBandwidthProof = 0x09
    MsgPayment        = 0x0A
    MsgMeshGossip     = 0x0B
    MsgBLEFragment    = 0x0C
    MsgGroupForm      = 0x0D
    MsgGroupKey       = 0x0E
)

type WireMessage struct {
    Version uint8
    Type    uint8
    Length  uint16
    Nonce   [12]byte
    Payload []byte
    MAC     [16]byte
}
```

B Relay Operator Requirements

1. Minimum 1000 PARS stake on L2
2. $\geq$ 95% uptime (rolling 30-day)
3. $\geq$ 10 Mbps upload
4. Public IP or reliable NAT traversal
5. Compliance with relay operator agreement

Table 7: Estimated Monthly Relay Rewards

Tier	**Traffic**	**Reward**
Small (10 Mbps)	500 GB	250 PARS
Medium (100 Mbps)	5 TB	2 500 PARS
Large (1 Gbps)	50 TB	25 000 PARS